Document Ingestion

Document ingestion is the process of receiving documents, validating them, extracting their content, cleaning the extracted text, and preparing the information for further processing.
Documents can come in different formats, such as PDF, DOCX, TXT, and scanned images. Before an AI application can use the information in these documents, it must make their content accessible in a consistent format.

Simple example: Imagine a company has 100 PDF files containing employee policies. An employee wants to ask an AI assistant, "How many annual leaves can I take?"
The AI application first needs to process those policy documents and make their contents searchable. Document ingestion prepares the information so that later components can find the relevant policy and use it to answer the question.

❖ Why is document ingestion important?

- **Makes information usable:** Converts raw documents into a format that software can understand and process.
- **Helps find information:** Organizes document content so the required information can be found easily.
- **Improves data quality:** Detects corrupted files, extraction mistakes, and incorrect data.
- **Supports automation:** Reduces the need for repetitive manual work when processing documents.
- **Supports AI applications:** Prepares document content for enterprise search, AI assistants, and RAG-based chatbots.
- **Tracks document details:** Stores information about document sources and processing steps, making it easier to track them.

Errors during document ingestion can affect every step that follows. For example, if a system reads the amount on a scanned invoice incorrectly, it may use the wrong amount in later calculations.

❖ Types of document ingestion

Document ingestion can be performed in three ways depending on how quickly documents need to be processed: batch, real-time, and micro-batch ingestion.

1. Batch Ingestion

Meaning: Documents are collected and processed together at scheduled intervals.
Real-world example: A bank processes all the day's transaction reports and account statements overnight to generate reports for the next morning.

2. Real-Time Ingestion

Meaning: Documents are processed as soon as they arrive, with minimal delay.
Real-world example: When a customer uploads documents for a loan application, the system immediately starts processing them so the application can move to the next step.

3. Micro-Batch Ingestion

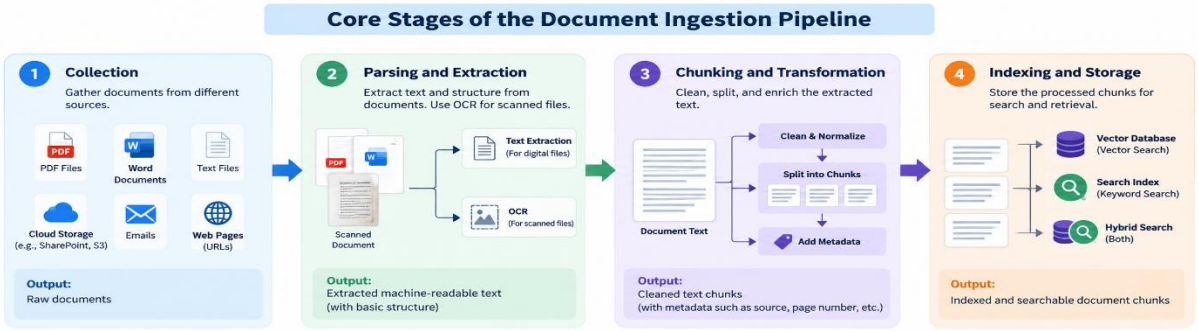
Meaning: Documents are collected in small groups over short intervals and then processed together.
Real-world example: An accounting system collects incoming invoice emails for every minute and processes the invoices together instead of processing each email separately.

Key Differences

	Batch	Real-Time	Micro-Batch
Processing	Large groups at scheduled times	Each document as it arrives	Small groups at short intervals
Speed	Slower; may take hours	Fastest response, usually seconds	Near real-time, often seconds to minutes
Best suited for	Historical reports and archives	Urgent applications and time-sensitive workflows	Invoice processing and email attachments

❖ Core Stages of the Document Ingestion Pipeline

A document ingestion pipeline consists of multiple stages. Each stage performs a specific task, and its output is passed to the next stage.



Stage 1: Document Collection

Documents are collected from different sources, such as:

- Local files and document repositories
- Cloud storage
- APIs and upload portals
- Emails and web pages

Output: Raw documents ready for processing.

Stage 2: Parsing and Data Extraction

The system reads documents and extracts information in a format that software can process.

- Text extraction: Extracts text from digital PDFs and Word documents.
- OCR (Optical Character Recognition): Reads text from scanned documents and images.
- Layout understanding: Identifies headings, columns, tables, and how different elements are arranged.

Output: Extracted text and information in a machine-readable format, while keeping the document structure wherever possible.

Stage 3: Chunking and Transformation

The extracted content is cleaned, organized, and prepared for further processing.

- Cleaning and normalization: Removes unwanted content and makes the formatting consistent.
- Chunking: Divides long documents into smaller sections.
- Metadata enrichment: Adds details such as the document title, source, page number, and section.
- Semantic chunking: Groups related information together instead of splitting text only by length.

Output: Smaller, organized sections of content with useful metadata.

Stage 4: Indexing and Storage

The processed content is stored and organized so that relevant information can be found when needed.

- Vector indexing: Helps find information based on meaning using embeddings.
- Keyword indexing: Helps find information by matching specific words or phrases.
- Hybrid search: Combines meaning-based search with keyword search.
- Storage: Saves documents, content sections, metadata, and indexes in suitable storage systems.

Output: Stored and searchable content that is ready for retrieval.

❖ How Document Ingestion Supports RAG

RAG stands for Retrieval-Augmented Generation. It is a technique that combines information retrieval with a Large Language Model (LLM) to generate answers using relevant information from external documents.

1. Document Collection
The process begins by collecting documents from sources such as PDF files, Word documents, company databases, cloud storage, and web pages. These documents contain the information that the AI application will use to answer questions.
2. Parsing and Content Extraction
The system extracts text and relevant structural information from the collected documents. If a document is scanned or contains text in an image, Optical Character Recognition (OCR) can be used to recognize the text.
This step converts the document content into a format that can be processed by the system.
3. Chunking and Transformation
The extracted content is cleaned and divided into smaller sections called chunks. Chunking allows the system to retrieve specific portions of a document instead of processing the entire document for every question.
Metadata, such as the document name, page number, or section title, can also be attached to help identify and organize the content.
4. Indexing and Storage
The prepared content is stored and organized for efficient retrieval. In a typical RAG implementation, the system may generate embeddings—numerical representations of text—and store them in a vector database to support semantic search. Other systems may use keyword search or a combination of search methods.
This step makes the document content available for retrieval when a user asks a question.
5. User Query and Information Retrieval
When a user submits a question, the RAG system searches the indexed content to identify the chunks that are most relevant to the question.
For example, if a user asks, "How many annual leave days are employees entitled to?", the system searches the available HR policy documents and retrieves the relevant section.
6. Answer Generation
The retrieved information is passed to a Large Language Model (LLM) along with the user's question. The LLM uses the retrieved context to formulate a relevant answer.
For example, if the retrieved HR policy states that employees receive 20 annual leave days per calendar year, the system can use that information to answer the user's question.
The answer can also include references to the original document, if the application supports source citations.

❖ Real-World Applications of Document Ingestion

AI assistants such as ChatGPT, DeepSeek, and Claude allow users to upload supported documents and ask questions about their contents.

Example: Suppose you upload a 50-page research paper and ask, "Summarize this paper and explain its main findings."

The process can be understood through the following steps:

1. **Document Collection:** You upload the research paper to the AI assistant.
2. **Parsing and Extraction:** The system extracts accessible text and relevant content from the document.
3. **Chunking and Transformation:** Depending on the system's implementation, the content may be divided into smaller sections and prepared for analysis.
4. **Retrieval or Context Selection:** Relevant information is identified and made available to answer your question.
5. **Answer Generation:** The AI generates a summary based on the information available to it.
6. **Result:** You can understand a lengthy research paper without having to read every page manually.

A practical implementation of Document Ingestion is available in the Jupyter Notebook linked below. The notebook demonstrates how documents are collected, processed, and prepared for AI applications, including extracting content, chunking, and organizing information for retrieval. You can access the project from this link: Document Ingestion – Jupyter Notebook